

THE STARSHIP GALILEO

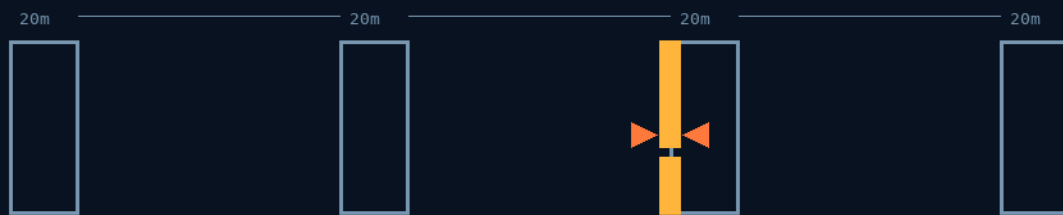
DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

PHASE 19

BULKHEADS: OPEN-SIDED INTERLOCKING HABITATION DECKS & HORIZONTAL BLAST GATES

Double-sided split armor — same as the outer hull
18-metre intervals — retract horizontal — symmetric deadbolts



POD POD POD

POD POD POD

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BULKHEAD DOORS EVERY 20 METRES, ROUGHLY — THE CORRIDOR IS 6 METRES

GP-IM-019

Revision v1.1 — 21 August 2026

20 of 290

VETTED: PASS — RAPID COMPARTMENT ISOLATION

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LICENSE: CERN OPEN HARDWARE LICENCE v2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 19

PHASE 19: BULKHEADS — OPEN-SIDED INTERLOCKING HABITATION DECKS & HORIZONTAL BLAST

GATES — STATUS: CURRENT. The ship's compartment spine: primary crew decks engineered with sliding, interlocking structural gates that snap shut horizontally to seal a localized hull breach without dropping pressure in the neighbouring sub-decks. The gates are hull-grade — double-sided split armor matching the Phase 13 plating spec — retracting horizontally into bulkhead pockets at 18-metre intervals, locked by two-way symmetric interlocking deadbolts. Vetted: PASS — rapid compartment isolation; the millisecond closure is demoted to TEST/SPECIFICATION and rides §9.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-019, v1.1 — 21 August 2026. The twentieth manual of the program — 20 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the bulkhead law as seated (contents line; the verbatim record — rhythm, gate spec, pod-row layout) — §2 why it works (pressure boundaries, graded) — §3 the system on the ship — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

VERSION NOTE (v1.0 — 20 August 2026): first build of this manual, on the Author's standing night-shift order (seated verbatim in sitting 499). The phase body stands exactly as seated in the master — split out of the combined PHASE 6, 8 & 19 cluster into its own standalone section in sitting 498 (the layout chronicle remains in the cluster as the sitting record; the manual carries the same blocks verbatim); no wording changed; the manual adds the assembly truth around it.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Engineers the primary crew decks with sliding, interlocking structural gates capable of snapping shut horizontally within milliseconds to seal off localized hull breaches

without dropping pressure in neighbouring sub-decks.' [VET CONSTRAINT, seated §5: the millisecond figure is TEST/SPECIFICATION, not certified — the bench owes the number.]

THE RHYTHM LAW (verbatim, the author's placement): 'bulkhead doors are every 20 metres roughly' — the rough law; the exact in-file figure is 18-metre intervals for the blast gates (below). Two nearby rhythms — doors at 18, the Phase 228 magnetosphere ring coils at 20 — live side by side; reconciling them is the author's ruling to speak (§9).

THE GATE SPEC (verbatim, from the vessel-scale block, in the file since the beginning): 'Segmented Blast Gates: Corridors feature double-sided split armor doors at 18-metre intervals that retract horizontally into bulkhead pockets Doors feature two-way symmetric interlocking deadbolts and match Phase 13 hull plating specs.'

THE POD-ROW LAYOUT (verbatim, same block): 'Upper Deck Cross-Section: Hull pod -> 3 Accommodation pods -> 6m corridor -> 2 public utility pods -> 6m corridor -> 12 accommodation pods -> 6m corridor -> 2 public utility pods -> 6m corridor -> 3 accommodation pods -> Hull pod. Accommodation cubes arranged in blocks of 6 (3 deep x 2 wide) with dedicated side access corridors.'

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE PHYSICS: this is straightforward compartmentalisation — the vet's own grade. The governing line: pressure differential + intact pressure boundary = force on the gate. A breach does not have to depressurise the ship if the pressure boundary can be subdivided; the gates subdivide it.

THE MILLISECOND CLAIM, DEMOTED: a large pressure door has mass, inertia, seals, locking mechanisms and a finite travel time — 'millisecond closure: TEST/SPECIFICATION, not yet proven' (the vet, seated). Rapid automatic isolation: PASS. The number rides the bench (§9).

THE PROJECTILE CAUTION, SEATED: the gate must be designed so that pressure differential neither prevents it closing nor turns the door into an uncontrolled projectile. Driven closure with retained guidance — the door is captive in its pocket track at all times.

THE ARMOR LOGIC: a blast gate that isn't hull-grade is a planned weak point in every compartment line — so the gates match the Phase 13 hull plating spec, seated. Double-sided split armor: the gate fights from both faces.

§3 — THE SYSTEM ON THE SHIP

- THE GRID: gates at 18-metre intervals on every corridor — ~28 gate lines per 500-metre deck run (25 if the author's 20-metre rhythm ruling supersedes — his to speak, §9).
- THE DOOR: double-sided split armor, retracts horizontally into the bulkhead pocket, two-way symmetric interlocking deadbolts, Phase 13 plating spec.

- THE FAILURE-MODE CHAIN (seated 21 August 2026, from the SITTING 425 vet — full record §5): the heavy-duty drives ride an internal slide rail PERPENDICULAR to door travel and in emergency detach and move sideways clear of the runner, so a dead drive never traps the door; if the centre lock jams, BOTH leaves slide one way and access comes via the open end; two mechanical winders (door face + adjacent room) release the drive-lock grips and move the drive with no power at all; pressure equalization (under-floor array to tanks) serves normal operation only — emergency access never waits on it.
- THE STATIONS: Phase 163 Bulkhead Wall Stations (pod medical kit, O2 breather bottles, CO2 extinguisher, both sides of the doors) recur at every gate line — the gate line is the ship's emergency furniture line.
- THE CORRIDORS: 6 metres wide, seated (the corridor-width fill-list item is CLOSED); pod rows per the verbatim cross-section — 96m of the 100m width accounted, the 4m remainder his call (§9).
- THE BRIDGE DOORS: Phase 51 puts the same Phase 19 split armor doors on the bridge evacuation conduit, with 5-second proximity auto-closure and deadbolt engagement.

INTEGRATION NOTE (audit, flagged): the ground-test ship confirmation rides the record — the Gemini set-3 conversation PDF confirms the ground-test ship uses 'the identical Phase 19 interlocking cube pods, 6-metre corridors, and 18-metre segmented horizontally retracting blast gates.' Earth-side testability is built in.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE VETERAN PAYOUT (seated law): 'Retirement at the 20-year marker entitles the veteran to 8 fully outfitted Phase 19 pods, transported to the destination of choice to instantly construct an interlocking estate or business module.' — the pod is a pension denomination.
- THE EMERGENCY GRID (seated doctrine): equipment is 'flush-mounted inside dedicated emergency recess fixtures on the interior cabin walls of every Phase 19/32 standardized pod' — the pod wall is a standard fitting plane.
- THE CARGO LINE (seated doctrine): un-pressurized Phase 19 cargo pods carry mining-grade prills (Phase 128) — the same cube, unpressurized, works freight.
- THE TOILET NESTING (seated arithmetic): toilets at the 100-metre rhythm land on every 5th bulkhead line under the 20m figure — under the exact 18m gate figure the nesting no longer lands clean; flagged, not forced (§9, his ruling).

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT, vet batch 9), VERBATIM GRADE: ' PASS — rapid compartment isolation; milliseconds requires testing.' Its physics concurrence is seated in §2 (pressure differential + intact boundary = force on the gate; no fundamental physics failure). Its demotion is seated with it: the millisecond closure claim is TEST/SPECIFICATION — mass, inertia, seals, locks, travel time — and its design caution is seated: pressure differential must neither prevent closure nor weaponise the door. The audit's own history is owned in the record: the door spec was never missing — the file calls them blast gates; the audit's grep was wrong, seventh confession, corrected same day (6 August 2026).

SECOND VET RECORD — SITTING 425 (18 AUGUST 2026), THE NEW BULKHEAD LOCKING SYSTEM, CARRIED INTO THIS MANUAL 21 AUGUST 2026 ON HIS NOTE (verbatim): "19 is missing the new bulkhead locking system from the vet" — HIS MECHANISM, VERBATIM FROM THE SITTING: "as these relate to bulkhead door with single pods pressure variance being of low danger component is either direction of vacuum and pressure. a per section under floor plumbing array per section to pressure tanks and the outer vacuum will equalize room pressures. the heavy duty mechanical drives of the bulkhead doors will be on an internal slide rail horizontally/perpendicular to the direct it operates. in an emergency, the drive mechanism will dethatch from the door inside the wall and will mechanically move sideways away from the door runner channel. Thus allowing the doors to be forced open. wherein the centre-lock is engaged and jammed both doors can simply be slid in one direction and access gained via the open end. where a power outage has occurred or control lines damaged. the built in mechanical winder in the door face will release the drive lock grips and the mechanical winder in the alternate room where the door slides"

THE CO-ENGINEER'S GRADE (SITTING 425, verbatim): '287 → PRE-VETTED MECHANICAL PRINCIPLE' — the previous 'invent emergency-opening mechanism' yellow flag retired. The audit's structure of his chain, seated: pressure differential runs EITHER direction (A over B, B over A, or vacuum one side), so the design never assumes a direction; the under-floor plumbing array equalizes sections for NORMAL operation, and equalization is deliberately NOT a requirement for emergency access; the drive physically clears the runner — 'once the powered/mechanical drive is physically removed from the load path, the door is no longer fighting the failed drive'; the centre-lock bypass removes the lock from the emergency-opening requirement entirely; the winders give a completely mechanical recovery chain under total power/control loss. WHAT REMAINS (ordinary engineering, owed to the bench, §9): size the transverse actuator slide for the door/pressure loads; ensure the drive cannot re-engage accidentally; positive mechanical indication it has cleared the runner; manual force calculated under the maximum specified pressure differential; the winders themselves accessible and never trapped by the same failure; physically test the complete sequence with pressure in BOTH directions.

§6 — BUILD SEQUENCE

- STEP 1 — Build the gate leaf: double-sided split armor to the Phase 13 plating spec (GP-IM-013 v1.4)
 - the leaf is a hull panel that moves, build it like one.
- STEP 2 — Build the pocket and track: horizontal retraction into the bulkhead pocket; the leaf is captive at all times — no free projectiles under differential (§2).
- STEP 3 — Fit the drive and the deadbolts: driven closure with two-way symmetric interlocking deadbolts; power-failure behaviour defined and benched (§9).
- STEP 4 — Set the grid: 18-metre intervals per the seated spec; gate lines carry the Phase 163 wall stations both sides; corridors at 6 metres.
- STEP 5 — Commission the closure logic: auto-close on breach detection, helm authority, local manual override; the Phase 51 bridge doors take the 5-second proximity auto-closure profile.
- STEP 6 — Bench the closure number (§9): mass, drive, seal, travel time — then and only then may a milliseconds figure be spoken.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 13 — the 94mm stack (GP-IM-013 v1.4): the gates match the hull plating spec by law; the stack's manual is the gate's materials reference.
- PHASE 228 — The Skeletal Magnetosphere: ring coils 'every 20 metres at a bulkhead' — the coil stations and the 18m gate lines are the two rhythms awaiting his ruling (§9).
- PHASE 163 — Bulkhead Wall Stations: the medical/O2/CO2 emergency furniture recurring at every gate line.
- PHASE 51 — the SMOT bridge evacuation conduit: same split armor doors, 5-second proximity auto-closure, deadbolt engagement.
- PHASE 6 & 8 — the 500m layout & 3D sliding matrix (GP-IM-006 v1.4): the pod grid the gates subdivide; the Clean Prism law the grid lives in.
- PHASE 32 — the standardized pod: the 19/32 pod wall is the standard fitting plane (§4).

§8 — DO-NOT-BUILD

- NEVER spec a gate below hull grade — a weak gate is a planned weak point in every compartment line (§2).
- NEVER certify milliseconds until the bench seats the number — TEST/SPECIFICATION stands (§5, §9).

- NEVER let differential jam or launch the door — captive track, driven closure, benched power-failure behaviour (§2, §6).
- NEVER move the rhythms unilaterally — 18 vs 20 is the author's ruling, not the builder's convenience (§9).

§9 — OWED

THE BENCH, OWED: the closure-time bench (leaf mass, drive authority, seal drag, travel time — the milliseconds number, if it survives); power-failure closure behaviour; deadbolt shear ratings; the 18m-gates vs 20m-coils rhythm ruling (his to speak); the 4m cross-section remainder (his call); toilet-line nesting under the 18m figure (flagged, not forced). All owed items ride the bench until spoken. *[PARTIALLY ANSWERED 21 AUGUST 2026 — 'power-failure closure behaviour' is now ANSWERED by the failure-mode chain (drive disengagement, centre-lock bypass, dual winders — §3/§5). The owed bench gains the SITTING 425 engineering list: transverse slide sizing; re-engagement lockout; positive cleared-runner indication; manual force at maximum differential; winder accessibility; the full sequence physically tested with pressure in both directions.]*

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-019, v1.1 — CURRENT, seated law, master v2.1 (numerical print order, post-split). The twentieth manual of the program — 20 of 290 delivered. Built 20 August 2026; v1.1 seated 21 August 2026 — the new bulkhead locking system carried in from the SITTING 425 vet on his note.

PROVENANCE — THE STANDING ORDER, VERBATIM (seated in sitting 499): '16 and then bed for me, after we finish that you can do 17 to 27 and i will read all 10 first thing to see if we can get through them faster in blocks'. The phase's own chain, all seated in the master: the vessel-scale block from the beginning (gate spec, pod-row layout); the 6 August 2026 layout sittings (the rhythm law, the blast-gate spec found, the corridor width found — with the audit's seventh confession owned); the naming batches of 12 August 2026 ('19: Bulkheads'); the vet batch 9 PASS; the split of 20 August 2026 that stood the phase alone (sitting 498, his order).

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete. Next version triggers: the §9 benches, the rhythm ruling, or his word. v1.1 (21 August 2026): the bulkhead locking system seated at §3 and §5; §9's power-failure item answered, the engineering list owed.

END OF MANUAL — PHASE 19